PHYSICAL CLIMATE RISK: FLOOD MODEL COMPARISON, HAZARD AND VULNERABILTY ASSESSMENT AND FINANCIAL IMPLICATIONS

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Master Thesis

NASLOV V SLOVENŠČINI

Magistrsko delo

Supervisor: Prof. Mitja Luštrek

Co-Supervisor: Prof. Boštjan Kaluža

Ljubljana, Slovenia, Month Year

Acknowledgments

Thank everyone who contributed to the thesis. Do not forget to mention any funding received.

Abstract

Flooding is one of the most frequent and damaging natural hazards globally, and its risk is projected to increase due to climate change. Accurate flood risk assessments are essential for spatial planning, climate adaptation, and ESG-aligned financial decision-making. This thesis investigates physical flood risk by comparing global and local hazard models, evaluating standardized vulnerability functions, and quantifying financial implications for real estate assets in Slovenia.

Two flood hazard models are assessed: the global WRI Aqueduct River Flood model and the high-resolution local model Integralna karta globin (IKG). A geospatial validation dataset comprising flood depth and damage records from four flood events in Slovenia (2010–2023) is used to evaluate model accuracy. Flood depths are analyzed using both regression and classification metrics, while vulnerability is assessed through comparison of observed damages against standardized depth–damage functions provided by the European Commission Joint Research Centre (JRC).

A key technical contribution is the development of a data processing pipeline that harmonizes heterogeneous IKG hazard data and integrates it into the modular Physrisk platform using the Zarr data format. This enables scalable, scenario-based risk assessment and model intercomparison.

Findings suggest that while local models offer improved accuracy for continuous flood depth prediction in regression tasks, the global model performs better in categorical classification. Proximity to water bodies was not found to significantly improve prediction accuracy. Depth–damage functions align well with observed damage distributions for residential properties, though segmentation by property type reveals variability in model performance.

The thesis contributes to the validation and integration of physical climate risk models for real estate applications and enables more accurate, location-specific flood risk estimates to support planning and investment decisions.

Povzetek

Povzetek v slovenščini naj ne bo daljši od ene strani.

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List of Algorithms

Abbreviations

JSI ... Jožef Stefan Institute
IPS ... International Postgraduate School
AEP ... Average Exceedance Probability
EAD ... Expected Annual Damage
ESG ... Environmental, Social, and Governance
JRC ... Joint Research Centre
RP ... Return Period
WRI ... World Resources Institute

Symbols

$j^\star$ ... black-body irradiance
σ ... Stefan's (or Stefan-Boltzmann) constant

Glossary

Glossary of terms

Chapter 1

Introduction

1.1 Background and Motivation

Flooding is a major natural hazard, responsible for substantial economic losses, social disruption, and loss of life across the globe. With changing climate conditions, flood risks are intensifying in both frequency and magnitude. The Intergovernmental Panel on Climate Change (IPCC) projects a rising likelihood of extreme hydrological events, including both pluvial and fluvial floods, due to higher precipitation variability and rising sea levels (TODO: REF).

In Europe, and particularly in Central and Eastern regions such as Slovenia, recent flood events have exposed vulnerabilities in both infrastructure and institutional response systems. The need for robust flood risk models that can inform spatial planning, insurance pricing, and asset management is therefore more pressing than ever. This is especially relevant for stakeholders in the real estate and financial sectors, who must now account for climate-related risks as part of their environmental, social, and governance (ESG) reporting and compliance requirements.

1.2 Climate Risk and ESG Disclosure

Climate risk refers to the potential adverse effects of climate change on economic systems, institutions, and physical assets. It is commonly classified into three interrelated categories: *physical*, *transition*, and *liability* risks [1].

- **Physical risks** arise from the direct consequences of climate variability and long-term climate change. These include acute hazards such as floods, storms, and wildfires, as well as chronic effects such as sea-level rise and shifting temperature trends. These risks can cause immediate physical damage and long-term degradation of asset value.

- **Transition risks** arise from the transition to a low-carbon economy. They include policy changes (e.g., carbon pricing, emissions regulations), technology disruption, evolving market preferences, and reputational risks associated with inaction or misalignment.

- **Liability risks** are considered by some [2] and encapsulate risks arising when those affected by anthoropogenic climate change seek compensation.

These categories are increasingly embedded in ESG-aligned regulatory frameworks such as those developed by the Task Force on Climate-related Financial Disclosures (TCFD)

and the Network for Greening the Financial System (NGFS). In the European context, the European Central Bank (ECB) requires significant institutions to assess, manage, and disclose climate-related and environmental risks, including physical risks. According to the ECB's supervisory guidance, banks are expected to integrate these risks into their internal governance, risk management, stress testing, and disclosure practices [3].

To operationalize physical risk in quantitative models, it is typically decomposed into three interacting components:

- **Hazard** – the probability and intensity of a climate-related event (e.g., flood depth for a 100-year return period),
- **Vulnerability** - the expected damage to an asset given the intensity of the hazard,
- **Exposure** - the presence and value of assets located in potentially affected areas.

These components form the foundation of physical climate risk models used in both regulatory compliance and voluntary disclosures. However, despite the wide adoption of this conceptual framework, practical implementation often suffers from limited model transparency, inconsistent spatial resolution, and poor validation—especially at the local scale relevant to property-level investment and risk management.

This thesis focuses specifically on **acute physical risks**, with a particular emphasis on riverine flooding and its financial impact on real estate assets. By comparing the performance of global and local flood models and evaluating standardized vulnerability functions against observed data, the work aims to contribute to more accurate and transparent risk quantification in support of ESG-aligned decision-making.

1.3 Problem Statement

While global flood hazard models such as the WRI Aqueduct River Floods provide broad coverage and climate scenario integration, they are often limited in spatial resolution and may not reflect local hydrological conditions. On the other hand, high-resolution local models, such as Slovenia's *Integralna karta globin* (IKG), offer detailed inundation mapping but are typically fragmented, static (i.e., not climate scenario-aware), and difficult to integrate into broader risk platforms.

Furthermore, flood vulnerability is typically modeled using generalized depth–damage functions derived from empirical or synthetic data. These functions may perform well in aggregate, but their applicability across different building types, geographies, and damage-reporting standards remains uncertain. This creates uncertainty in downstream financial estimates and limits the usefulness of such models for asset-level ESG disclosure.

1.4 Research Objectives

The purpose of this thesis is to evaluate how well existing flood models can represent real-world flood depths and damage outcomes in Slovenia. Specifically, it compares a global hazard model (WRI Aqueduct) and a high-resolution local model (IKG) to assess the accuracy of their flood depth predictions.

Separately, the thesis tests global depth–damage functions, used to estimate flood-related property value loss, against reported damage data, with the goal of evaluating and potentially improving their performance. Together, these analyses aim to assess the reliability of current modeling tools for understanding physical climate risk and supporting property related decision making in the context of increasing flood exposure.

The main goals of the research are:

1. To improve existing physrisk risk assessment framework with the following contributions:
   (a) Develop a data processing pipeline for creating a unified and homogenous SI IKG flood depth model for different areas of importance.
   (b) Develop a data onboarding pipeline for integrating this new unified model into physrisk framework.
2. To compare the performance of a global flood hazard model (WRI Aqueduct) and a local Slovenian model (IKG) using real flood data.
3. To evaluate how model resolution and proximity to water bodies affect prediction accuracy.
4. To assess the accuracy of global flood depth–damage functions on reported damage cases in Slovenia and improve their performance.

1.5 Methodological Approach

The thesis adopts a comparative modeling approach using geospatial and statistical techniques. A validation dataset, compiled from the records of the Slovenian Administration for Civil Protection and Disaster Relief (*Uprava za zaščito in reševanje*), includes appraised damage and measured flood depths for selected events between 2010 and 2023.

Flood hazard models are evaluated using both regression-based and classification-based metrics to assess depth prediction accuracy. Vulnerability models are tested by comparing observed versus predicted damages using segmented analysis by flood depth and real estate type. All data are processed and harmonized into a common geospatial reference system and stored in Zarr format, enabling scalable access within the *Physrisk* platform.

From a scientific perspective, this work contributes to the empirical validation of global flood hazard and vulnerability models in a local Central European context. From a practical standpoint, it improves the transparency and reusability of flood risk data pipelines, supporting efforts to build ESG-compliant risk disclosure tools.

1.6 Structure of the Thesis

The remainder of the thesis is structured as follows:

- **Chapter 2** presents a review of relevant literature and related work on flood hazard modeling, vulnerability estimation, and ESG-aligned risk frameworks.
- **Chapter 3** describes the study area, datasets, and the methodological framework used for model evaluation and data processing.
- **Chapter 4** details the integration of the local IKG model into the Physrisk framework and presents the architecture of the comparison toolset.
- **Chapter 5** provides experimental results, including quantitative evaluation of model performance, depth–damage function analysis, and financial impact estimates.
- **Chapter 6** discusses the key findings, limitations, and possible future research directions, and concludes the thesis.

Chapter 2

Literature Review & Related Work

TBD

Chapter 3

Study Area, Data, and Methodology

This chapter defines the geographic and empirical scope of the study and outlines the data sources and methodological approaches used to evaluate flood hazard and vulnerability models. The objective is to establish a clear foundation for the experimental work presented in later chapters by describing the spatial context, data quality, and analytical framework adopted in this thesis.

First, the chapter introduces the two primary flood hazard models under investigation: the global WRI Aqueduct River Flood model and the local Slovenian Integralna karta globin (IKG) model. These models are compared in terms of resolution, coverage, return period granularity, and underlying assumptions.

Second, the chapter presents the vulnerability component of the flood risk framework, focusing on standardized depth–damage functions developed by the European Commission Joint Research Centre (JRC). These functions are used to estimate expected economic damage from flooding based on inundation depth and building type.

Lastly, the chapter describes the overall flood risk modeling approach, structured according to the widely adopted hazard–vulnerability–exposure paradigm. This includes the assumptions used for data transformation, the structure of the validation dataset, and the rationale behind the chosen evaluation metrics.

The chapter is structured as follows:

- Section 3.1 describes the hazard models and their flood depth data characteristics.
- Section 3.2 presents the vulnerability modeling approach using depth–damage functions.
- Section 3.3 integrates hazard, vulnerability, and exposure into a unified flood risk modeling framework.

3.1 Hazard and Depth

Flood hazard refers to the likelihood and severity of flooding at a specific location over a defined time horizon. In this thesis, hazard is represented by flood depth estimates associated with given return periods (e.g., 10-, 100-, or 500-year floods). These estimates form the input to downstream risk calculations and serve as the basis for comparing model performance.

This section describes the two flood hazard models used in the study: the global WRI Aqueduct River Flood model and the local Slovenian Integralna karta globin (IKG) model. The models differ significantly in spatial resolution, geographic coverage, underlying hy-

drological assumptions, and treatment of climate scenarios. Their characteristics are summarized in Table 3.1.

3.1.1 WRI Aqueduct River Flood Model

The *World Resources Institute (WRI) Aqueduct River Flood* model is a publicly available global flood hazard dataset that provides inundation depth estimates at a spatial resolution of approximately 1 km. It is designed for large-scale screening of riverine flood risks and supports a wide range of return periods, typically including RP2, RP5, RP10, RP25, RP50, RP100, RP250, RP500, and RP1000.

The model incorporates climate scenario data by linking river discharge estimates to Representative Concentration Pathways (RCPs), particularly RCP4.5 and RCP8.5. It uses a combination of global hydrological models and digital elevation data to simulate flood extents and depths under historical and future climate conditions.

In this thesis, the WRI model is used as a benchmark for evaluating local model performance under the 100-year return period (RP100). While it lacks the spatial granularity needed for parcel-level risk assessment, it offers consistent global coverage and climate-adjusted projections.

3.1.2 Integralna karta globin (IKG)

The *Integralna karta globin (IKG)* is a collection of high-resolution flood hazard models developed for selected municipalities in Slovenia. Each model is based on detailed hydrological and hydraulic simulations conducted by local engineering firms, using fine-scale terrain data and historical flood records. The original data is stored in vector formats (e.g., shapefiles, GeoPackages), with flood depth zones categorized into three discrete classes:

- GM: Minor inundation (0.5 m depth)
- GS: Significant inundation (1.5 m depth)
- GB: Severe inundation (2.5 m depth)

IKG models are available for return periods of 10, 100, and 500 years (RP10, RP100, RP500) and are typically generated in EPSG:3794 projection. For this thesis, the IKG layers from 18 areas of importance (*Območja pomembnega vpliva*) were harmonized, rasterized into a 5×5 m grid, and integrated into a unified hazard layer (as described in Chapter 4).

Although IKG models are not climate-scenario-aware, they are more spatially detailed than global models and are considered to better represent flood dynamics at the neighborhood or parcel level.

3.1.3 Return Periods and Depth Representation

Both hazard models use a return-period-based approach to represent flood hazard. A return period T indicates the average time between flood events of a given magnitude and is used to estimate the annual exceedance probability $p = 1/T$.

Flood depths are modeled either as continuous variables (WRI) or discrete depth classes (IKG). For consistent comparison, the IKG model was converted into a raster with fixed depth values for each class. For WRI, continuous flood depths are directly extracted for the RP100 layer and compared against observed values.

Table 3.1: Comparison of flood hazard models used in the study

Feature	**WRI Aqueduct River Flood**	**Integralna karta globin (IKG)**
Spatial resolution	∼1 km	5 m
Coverage	Global	18 regions in Slovenia
Climate scenarios	RCP-aware (e.g., RCP4.5, RCP8.5)	Not scenario-aware
Depth type	Continuous (in meters)	Discrete classes (0.5, 1.5, 2.5 m)
Return periods	2 to 1000 years	10, 100, 500 years
File format	Raster (GeoTIFF)	Vector (shapefile, GPKG)
Coordinate system	EPSG:4326	EPSG:3794 (transformed)

3.1.4 Hazard Data Extraction

For model comparison and validation, flood depth predictions from both WRI and IKG were extracted for each point in the validation dataset corresponding to flood events in Slovenia between 2010 and 2023. All extractions were performed for the 100-year return period (RP100) using the toolset described in Section 4.4. The extracted depths were then compared to observed values using regression and classification metrics, as discussed in Chapter 5.

This dual-model approach enables an empirical assessment of the trade-offs between resolution and scalability and provides insight into the suitability of global models for local decision-making.

3.2 Vulnerability and Damage Function

Vulnerability in the context of flood risk refers to the degree of damage an asset is expected to incur when exposed to a flood of a given intensity. In quantitative flood risk models, vulnerability is typically represented by a *depth–damage function* that expresses the expected percentage of asset damage as a function of inundation depth.

This section describes the vulnerability modeling approach adopted in this thesis. It focuses on the use of standardized global depth–damage functions, their assumptions, and how they are applied to the Slovenian flood impact dataset. The function used here originates from the Joint Research Centre (JRC) of the European Commission and is part of a global initiative to provide harmonized vulnerability data for risk assessment.

3.2.1 Depth–Damage Functions

A depth–damage function (DDF) maps flood depth h (in meters) to expected relative damage $D(h)$, typically expressed as a fraction or percentage of the total asset value. These functions are generally derived from empirical post-disaster surveys, insurance claims data, or synthetic modelling. They vary by asset type, construction material, country, and socio-economic context.

In this study, we use the depth–damage functions published by Huizinga et al. [4], which are among the most widely used sources for flood damage estimation at the European level. These functions are defined for different building types (e.g., residential, commercial, industrial) and are based on a normalized building replacement value (BRV).

3.2.2 Function Structure and Properties

The JRC functions have the following properties:

- They are monotonic and piecewise continuous,
- Damage begins at a threshold of approximately 0.1–0.2 m and increases with depth,
- Maximum damage is capped between 70% and 90% depending on asset type,
- Uncertainty is not explicitly modeled, but curves represent expected mean damage.

An example residential depth–damage function is illustrated in Figure **??**. In this model, a 0.5 m flood may cause damage equivalent to 15–25% of the building's replacement value, while a 2.5 m flood could exceed 70%.

3.2.3 Application to Slovenian Flood Events

To evaluate the applicability of global depth–damage functions in a local context, we apply the JRC DDFs to a dataset of reported flood damages in Slovenia. This dataset, obtained from the Administration for Civil Protection and Disaster Relief, contains:

- Georeferenced flood locations,
- Reported flood depths (rounded or estimated by appraisal),
- Reported economic damage (in EUR),
- Property classification (e.g., residential, commercial, public use).

The reported damages are normalized by the building replacement value (when available or estimated), resulting in a relative damage metric. Predicted damage from the DDF is then compared to observed relative damage to assess model performance. The comparison is segmented by property type and flood depth category.

3.2.4 Assumptions and Limitations

While the JRC depth–damage functions offer a standardized approach, several assumptions and limitations must be noted:

- **Transferability:** The functions are designed for generalized application and may not reflect local building codes, materials, or insurance systems in Slovenia.
- **No uncertainty modeling:** The functions provide point estimates of expected damage, not distributions. However, empirical data show that actual damages are often bimodal or skewed.
- **Property segmentation:** Limited metadata in the Slovenian dataset constrains the ability to differentiate vulnerability by building age, elevation, or structural resilience.

Despite these limitations, the DDFs serve as a useful benchmark for evaluating vulnerability model performance. Further improvements are discussed in Section 5.4 and Chapter 6.

3.2.5 Link to Risk Estimation

The vulnerability model is used in conjunction with flood depth predictions from hazard models to compute expected damage at each location:

$$\text{Expected Damage} = D(h) \times \text{Asset Value}$$

This formulation assumes deterministic vulnerability, where the same flood depth always results in the same relative damage. This simplification facilitates large-scale analysis but may under-represent epistemic and aleatory uncertainty. These limitations are acknowledged in the discussion.

In the next section, we present the overall structure of the flood risk model used in this study and describe how hazard, vulnerability, and exposure components are integrated.

3.3 Flood Risk Modeling

Flood risk is commonly conceptualized as the intersection of three key components: hazard, vulnerability, and exposure. This approach is consistent with methodologies used in climate risk disclosure frameworks such as the Network for Greening the Financial System (NGFS), the EU Taxonomy, and the OS-Climate Physrisk platform. In this section, we formalize how these components are operationalized in this thesis to estimate physical flood risk at the property level.

3.3.1 Risk Definition

Following the standard formulation, flood risk R is defined as:

$$R = H \times V \times E$$

where:

- H is the hazard, represented by the probability and intensity (depth) of flooding for a given return period,
- V is the vulnerability, expressed as a depth–damage function mapping flood depth to expected relative damage,
- E is the exposure, typically the asset value (e.g., building replacement value) at the location in question.

In this formulation, risk can be expressed in monetary terms as the expected annual damage (EAD) or scenario-specific loss (e.g., loss under a 100-year flood). This thesis focuses on evaluating model performance for the RP100 return period.

3.3.2 Hazard Component

As described in Section 3.1, the hazard component is derived from either the global WRI Aqueduct River Flood model or the local IKG model. For each validation point, flood depth predictions are extracted at RP100 and used as input to the vulnerability function.

In areas where both models are available, their predictions are compared side-by-side using observed flood depths from past events as a reference. The evaluation metrics include regression error, classification accuracy across depth bands, and distance from reported measurements.

3.3.3 Vulnerability Component

As detailed in Section 3.2, the vulnerability component uses standardized depth–damage functions developed by the JRC. These functions convert predicted flood depths into expected relative damage fractions. The predicted damage values are then compared against reported economic losses in Slovenia, normalized by estimated asset values.

3.3.4 Exposure Component

The exposure term E is represented by the estimated or reported value of the affected asset. In the validation dataset, appraised building values are used when available; otherwise, standardized assumptions based on national averages are applied. While this introduces potential error, the relative damage metric allows for comparison independent of absolute valuation accuracy.

3.3.5 Implementation Workflow

The computational steps for flood risk estimation are as follows:

1. For each point in the validation dataset:
   - Extract flood depth from WRI and IKG models for RP100,
   - Apply the JRC depth–damage function to compute expected relative damage,
   - Multiply by estimated asset value to obtain monetary loss estimate (if applicable).
2. Compare model predictions (depth and damage) with observed values from reported flood events.
3. Segment results by model type, proximity to water bodies, and property classification.

This workflow is visualized in Figure **??**, illustrating the flow from geospatial hazard data to quantitative risk outputs.

3.3.6 Use Cases and Relevance

The flood risk modeling framework developed in this thesis supports a range of applications, including:

- Empirical evaluation of model accuracy for flood depth and damage estimation,
- Validation of global vulnerability functions in local contexts,
- Comparative analysis between coarse-resolution and high-resolution hazard models,
- Integration into ESG-aligned climate risk disclosure pipelines.

This model architecture is aligned with the structure of the Physrisk platform and is used in subsequent chapters to evaluate the performance of hazard and vulnerability models on observed flood events in Slovenia.

Chapter 4

Model Integration and Analysis Framework

Accurate flood risk assessment requires the integration of diverse data sources and computational tools to simulate hazard intensity, estimate potential damages, and evaluate financial exposure. In this chapter, we present the technical framework developed to combine heterogeneous flood hazard data—particularly local Slovenian models—into a unified system suitable for analysis and comparison within the Physrisk platform.

The focus lies on harmonizing the *Integralna karta globin* (IKG) flood maps, which vary in structure, format, and coordinate reference systems, and transforming them into a consistent, machine-readable representation. These models are rasterized, standardized by return period, and encoded in the cloud-optimized Zarr format. The processed datasets are then onboarded into the modular Physrisk platform, which separates hazard, vulnerability, and financial components and allows for flexible model evaluation.

To facilitate empirical comparison, the framework also includes a toolset that enables side-by-side querying and batch evaluation of both global and local models at precise geolocations, aligned with the validation dataset. This infrastructure supports all subsequent experimental analysis presented in Chapter 5 and lays the groundwork for scalable, reproducible climate risk assessments.

The chapter is organized as follows:

- Section 4.1 presents the overall system architecture and the interaction between components.
- Section 4.2 describes the pipeline for transforming and combining IKG models across multiple Slovenian regions.
- Section 4.3 explains how the standardized hazard data are integrated into the Physrisk platform.
- Section 4.4 outlines the model comparison toolset used to evaluate prediction accuracy across models.

4.1 Architectural Diagram

The integration of global and local flood hazard models into a unified analysis platform requires a modular, extensible architecture that supports heterogeneous input formats, scenario-aware modeling, and scalable evaluation. This section presents the architecture developed for this purpose, highlighting its components, data flow, and interoperability within the *Physrisk* framework.

4.1.1 System Overview

Figure **??** illustrates the architecture of the flood hazard model integration pipeline, based on the open-source *OS-Climate Physrisk* ecosystem. The process begins with raw flood hazard data, which may originate from global sources such as the WRI Aqueduct or TU Delft flood models, or from local sources such as the Integralna karta globin (IKG) used in Slovenia. These datasets are onboarded into the `os-risk-hazard` module, a Python-based processing library designed to standardize, harmonize, and prepare hazard indicators for risk evaluation.

The onboarded models are then encoded in Zarr format and stored in an AWS S3-compatible object storage. The Physrisk framework retrieves hazard indicators from this backend and combines them with vulnerability and exposure components to compute impact distributions. The resulting outputs can be accessed via the Physrisk API or used directly for ESG reporting and financial risk analysis through the `os-risk-financial` module.

4.1.2 Data Flow and Model Interoperability

Each flood model—whether global or local—is processed through the same onboarding interface, allowing standardization of outputs while preserving return-period-specific intensity information. The hazard models are structured into three-dimensional Zarr arrays indexed by latitude, longitude, and return period. This enables consistent querying and evaluation across models with different spatial resolutions or geographic coverage.

Once onboarded, all hazard layers become interchangeable within the Physrisk pipeline. Vulnerability models (e.g., depth–damage functions) and exposure datasets (e.g., building values or asset locations) can then be used in combination with any hazard source. This interoperability facilitates both comparative experiments and compound risk evaluations, where multiple models are used to bound uncertainty or stress-test assumptions.

4.1.3 Integration of the IKG Model

The locally developed IKG model, originally provided in shapefile and GeoPackage vector formats, required a dedicated pre-processing stage to align with the Physrisk architecture. This process involved:

- Transforming coordinate systems (from EPSG:3794 to EPSG:4326),
- Rasterizing vector layers into a fixed-resolution 5x5m grid,
- Assigning discrete flood depth values to each return period (RP10, RP100, RP500),
- Converting the raster layers into Zarr format for chunked cloud storage.

The unified model was then uploaded to AWS S3 and integrated into the Physrisk hazard module alongside WRI and TU Delft models. This enabled direct comparison and batch evaluation of global versus local model predictions, described further in Section 4.4.

4.1.4 Hazard Onboarding Pipeline

The transformation from raw IKG data into a cloud-optimized flood hazard model is summarized in Figure **??**. The pipeline consists of the following steps:

1. **Rasterization:** Converting input vector data into gridded raster tiles using a 5x5m resolution grid.

2. **Zarrify:** Chunking the raster into 1000x1000 tiles and encoding into Zarr format.
3. **Upload:** Transferring the data to AWS S3 for use within Physrisk.

The final onboarded dataset can be queried via the Physrisk API or accessed directly for integration with ESG tools or batch model evaluation.

4.2 Pipeline for Combining Heterogeneous IKG Models

The *Integralna karta globin* (IKG) flood hazard datasets are developed for individual municipalities or regions in Slovenia, each based on local hydrological and hydraulic analyses. These datasets differ in file structure, geospatial reference systems, data quality, and encoding of flood depths. To integrate these into a unified flood hazard model suitable for comparative evaluation and platform onboarding, a standardized processing pipeline was developed.

4.2.1 Data Sources and Heterogeneity

The raw IKG datasets were obtained from multiple engineering consultancies (*Inženiring za vode d.o.o., Urbania d.o.o., iS Projekt d.o.o., IZVOR d.o.o., DHD d.o.o.*), each responsible for flood modeling in a subset of areas classified as "Območja pomembnega vpliva" (Areas of Significant Flood Risk) by Slovenian water authorities.

Each local model typically includes:

- GIS vector data (shapefiles, GeoPackages) representing inundation zones,
- Attribute data encoding flood extent and classes for selected return periods (e.g., 10-, 100-, and 500-year events),
- Region-specific coordinate systems (typically EPSG:3794, Slovenia National Grid).

In total, data from 18 areas of importance were collected, with each area providing flood scenarios for three return periods. This resulted in 54 distinct layers that needed to be harmonized.

4.2.2 Standardization and Processing Workflow

To unify the IKG datasets into a single model, the following pipeline was developed and implemented in Python using open-source geospatial processing libraries (GDAL, Fiona, Rasterio, xarray):

1. **Coordinate Transformation:** All source files were transformed from EPSG:3794 to WGS84 (EPSG:4326) to ensure compatibility with global models and standardized grid alignment.
2. **Rasterization:** Vector layers were rasterized into a regular grid with a resolution of 5 meters (5x5m), which approximates the native resolution of the original hydrological models. A fixed flood depth value was assigned to each flood class:
    - GM (minor flooding): 0.5 m
    - GS (significant flooding): 1.5 m
    - GB (severe flooding): 2.5 m

3. **Return Period Alignment:** Each rasterized layer was tagged with its associated return period (RP10, RP100, RP500). For each area, three separate rasters were produced and stacked by return period.

4. **Tiling and Merging:** The processed rasters from all areas were mosaicked into a common spatial grid and stored as a 3D array indexed by latitude, longitude, and return period. Conflicts in overlapping regions (e.g., model boundaries) were resolved by assigning priority to higher-confidence datasets based on metadata quality.

4.2.3 Unified Flood Hazard Product

The final result of this pipeline is a unified flood hazard model for Slovenia covering all 18 areas of importance. This model:

- Captures flood depth estimates at high spatial resolution for three return periods,
- Standardizes input data from diverse local models into a single harmonized format,
- Enables consistent comparison with global models such as WRI Aqueduct within the same framework.

Figure **??** provides a schematic representation of the transformation process, from raw shapefiles to a standardized hazard product.

This unified model was subsequently converted to Zarr format and onboarded into the Physrisk platform, as described in Section 4.3.

4.3 Physrisk Framework – Model Onboarding

Once the local IKG flood models were unified into a standardized and spatially consistent format, the next step involved integrating them into the modular *Physrisk* framework. Physrisk is part of the broader OS-Climate open-source ecosystem and is designed to facilitate scalable, scenario-based physical climate risk analysis. The framework separates hazard, vulnerability, and financial components and enables flexible model composition, comparison, and evaluation.

4.3.1 Physrisk Architecture and Modules

The Physrisk architecture is composed of several interoperable modules:

- **OS-risk-hazard**: responsible for ingesting and formatting hazard data sets (e.g., flood depth by return period) into a common format.
- **OS-risk-physrisk**: computes impact distributions by convolving hazard data with vulnerability functions and exposure information.
- **OS-risk-financial**: translates physical impacts into financial metrics such as loss of asset value or disruption to operations.
- **Physrisk API**: exposes model outputs through a standardized programmatic interface.

The onboarding of hazard models occurs within the `os-risk-hazard` component and requires the hazard data to conform to a Zarr-based storage structure optimized for parallel access and spatial querying.

4.3.2 Zarr Format and Indexing

The Zarr format is a cloud-native format for storing large multi-dimensional arrays. It was chosen for hazard model storage due to its ability to:

- Efficiently compress and store gridded data in chunks (e.g., 1000×1000 tiles),
- Allow selective download of spatial subsets,
- Support multi-indexing (e.g., latitude, longitude, return period).

For the unified IKG model, the hazard data were encoded into a 3-dimensional Zarr array with the following dimensions:

- i: Latitude (y-axis),
- j: Longitude (x-axis),
- k: Return period index, corresponding to RP10, RP100, and RP500.

Each tile was stored as a compressed chunk and uploaded to an AWS S3-compatible bucket accessible to all Physrisk modules.

4.3.3 Onboarding Procedure

The onboarding process was implemented using the Physrisk hazard onboarding tool, a Python-based utility designed for structured hazard model integration. The steps were as follows:

1. **Prepare Metadata:** Define the return periods, CRS (EPSG:4326), spatial resolution, and transformation matrix for the IKG model.
2. **Convert to Zarr:** Use the onboarding tool to encode the processed GeoTIFF files into a Zarr store, assigning appropriate chunk sizes (e.g., 1000×1000 pixels).
3. **Upload to Cloud Storage:** Transfer the Zarr directory to a cloud-accessible S3 bucket used by the OS-Climate infrastructure.
4. **Register the Model:** Add a model reference entry into the hazard configuration of Physrisk, enabling downstream access via API and batch processors.

4.3.4 Integration with Other Models

With the IKG model onboarded, it became available in the same infrastructure alongside global flood hazard layers, including:

- WRI Aqueduct River Flood (global, 1 km resolution),
- TU Delft River Flood (Pan-European, 100 m resolution).

This interoperability allows any of the onboarded hazard layers to be used with common vulnerability functions and exposure datasets, enabling direct comparisons, uncertainty bounding, and ensemble evaluations. The IKG model is now fully accessible via the Physrisk API and can be used for property-level flood risk analysis in Slovenia under a unified modeling interface.

4.4 Comparison Toolset

To evaluate and compare the predictive performance of flood hazard models, a model-agnostic comparison toolset was developed and integrated into the Physrisk framework. This toolset enables side-by-side querying of multiple hazard sources at specific geolocations and return periods, aligned with the reference flood observations used for validation in Chapter 5.

4.4.1 Purpose and Design Principles

The comparison toolset was designed with the following goals in mind:

- **Model-agnostic access:** Allow querying of different onboarded hazard models (e.g., IKG, WRI Aqueduct) through a unified interface.
- **Return-period specificity:** Ensure comparability by extracting flood depth predictions at the same return period (e.g., 100-year flood).
- **Location-accurate evaluation:** Retrieve model outputs at high spatial precision to match reported flood locations.
- **Batch processing support:** Enable evaluation across hundreds of observations using efficient geospatial access patterns.

4.4.2 Interface and Functionality

The toolset consists of a lightweight Python interface built on top of the Physrisk hazard API. Given a set of coordinates (latitude, longitude) and a desired return period, the tool performs the following steps:

1. **Coordinate Transformation:** Convert input coordinates into the grid index of the onboarded Zarr array using affine transformations defined in the model metadata.
2. **Chunk Identification and Retrieval:** Locate the Zarr chunk containing the requested point and load the corresponding depth value for the specified return period.
3. **Output Formatting:** Return the flood depth value, return period, and model metadata for downstream use in regression or classification analysis.

When applied in batch mode, the tool leverages parallel reads and lazy loading to evaluate large datasets (e.g., the full validation set of flood reports) with minimal memory overhead.

4.4.3 Supported Models and Inputs

The toolset was validated for use with the following onboarded hazard models:

- **SI IKG (Unified Model)**: High-resolution local flood model rasterized at 5x5 m, covering 18 Slovenian areas of importance, with RP10, RP100, and RP500 layers.
- **WRI Aqueduct Flood Hazard**: Global model with 1 km resolution, scenario-aware, supporting multiple return periods from RP2 to RP1000.

The input to the toolset is a tabular dataset containing:

- Latitude and longitude of each observation,
- Desired return period (e.g., RP100),
- Optional fields: timestamp, municipality, reported depth, reported damage.

4.4.4 Integration with Validation Workflow

The comparison outputs are stored in a structured table, allowing for direct computation of model performance metrics such as:

- **Regression error:** difference between predicted and observed flood depths,
- **Classification accuracy:** match rate within flood depth categories (e.g., <0.5 m, 0.5–1.5 m, >1.5 m),
- **Bias analysis:** mean and median error segmented by proximity to water, terrain type, or municipality.

These outputs form the quantitative basis for the experimental results presented in Chapter 5.

4.4.5 Reproducibility and Extensibility

The toolset was developed as a standalone module and version-controlled as part of the project's repository. This design ensures that future updates to hazard models (e.g., inclusion of climate scenario variants or additional regions) can be incorporated without changes to the validation logic.

Furthermore, the modularity of the toolset supports extension to:

- Additional hazard types (e.g., pluvial flooding),
- Vulnerability assessments (via model chaining),
- Scenario-based risk projections (by switching hazard layers).

Thus, the comparison toolset serves not only as a core utility for this thesis, but also as a scalable evaluation framework for broader applications in physical climate risk modeling.

Chapter 5

Experimental Design and Evaluation

TBD

5.1 Experimental Goals

TBD

5.2 Experimental Setup

TBD

5.3 Experimental Results: Flood Depth

TBD

5.4 Experimental Results: Damage Functions

TBD

5.5 Analysis of Impact on Total Exposed Property Value

TBD

Chapter 6

Conclusions & Discussion

6.1 Conclusion & Interpretation of Results

TBD

6.2 Limitations of This Work

TBD

6.3 Future Work

TBD

Appendix A

Proofs of Theorems

A.1 Proof of the Pythagorean Theorem

Let us prove the Pythagorean Theorem from page **??**.

Proof. This proof is based on the proportionality of the sides of two similar triangles, that is, upon the fact that the ratio of any two corresponding sides of similar triangles is the same regardless of the size of the triangles.

Let ABC represent a right triangle, with the right angle located at C, as shown in Figure A.1. We draw the altitude from point C, and call H its intersection with the hypotenuse AB. Point H divides the length of the hypotenuse into two parts.

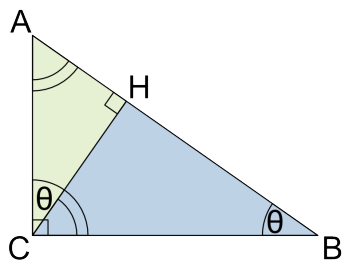

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem.

The new triangle ACH is similar to triangle ABC, because they both have a right angle (by definition of the altitude), and they share the angle at A, meaning that the third angle will be the same in both triangles as well, marked as θ in Figure A.1. By a similar reasoning, the triangle CBH is also similar to ABC.

Similarity of the triangles leads to the equality of ratios of corresponding sides:

$$\frac{BC}{AB} = \frac{BH}{BC} \text{ and } \frac{AC}{AB} = \frac{AH}{AC}. \tag{A.1}$$

The first result equates $\cos\theta$ and the second result equates $\sin\theta$.

These ratios can be written as:

$$BC^2 = AB \times BH \text{ and } AC^2 = AB \times AH. \tag{A.2}$$

Summing these two equalities, we obtain:

$$BC^2 + AC^2 = AB \times BH + AB \times AH = AB \times (AH + BH) = AB^2, \tag{A.3}$$

which, tidying up, is the Pythagorean theorem:

$$BC^2 + AC^2 = AB^2. \tag{A.4}$$

□

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Bibliography

Publications Related to the Thesis

All publications related to the thesis should be referenced in the text.

Other Publications (optional)

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Biography

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